

throughput. The true per pin architecture in this system makes all the analogue and digital resources always available on each system channel.

Multi-core testing and higher throughput

With the increased competition and demanding markets, it is certain that PCB OEMs and the other electronic manufacturing services need to boost their throughput and yield.

"In India, labour costs are increasing day by day and the cost of space has also increased over the last few years. So, if you use one operator, one fixture, one PC, you can save your cost as well as space. And mathematically speaking, previously we were having one tester and one fixture but now we are having two testers or more in one fixture, thus the throughput is increased by at least two times," says Ravi Richaria, manager sales & service support, Test & Research India Pvt Ltd.

The newer parallel and multi-core testing architectures available from vendors solve this problem. These parallel test architecture ICT systems deliver true multi-core test capabilities to test multiple units concurrently, thereby eliminating test time bottlenecks and increasing your testing productivity as compared to the previous ICT systems. Using the multi-core test systems minimises your system size and cost as these utilise common cooling, power and other infrastructural elements. In addition to system size and density, smaller physical configurations and short signal paths help to deliver high parallel test efficiency.

Integrated test methodologies

Testing of current complex circuit boards requires separate analogue, digital and other test workflows. So each test workflow accounts for individual test systems, multiple test fixtures, and separate operators. As circuit boards become more advanced, the traditional ICT technologies are becoming less effective in achieving reliable test coverage. The advent of SMT IC technolo-

gies with IC packages like BGA ball grid array (BGA) has made it difficult to directly access the pins using the traditional 'bed of nails' tester.

To solve this, ICT systems now integrate multiple test methods, where some points make contact through bed of nails and the others use edge connectors or JTAG ports or in-circuit emulators, thus using three or four test methods in parallel.

Ganapathy from QmaxTest says, "You should not depend too much on one test methodology or a particular approach. Integrated ICT solutions are a boon. Combinational approach has an advantage of two or more techniques so that yield is more and efficiency is increased."

TR5001T SII TINY from Test & Research India is the compact version with just 640 test points but it supports extended coverage with boundary scan in a compact and affordable solution. This desktop ICT offers parallel testing using multiple USB-connected units, audio analyser and data acquisition modules.

Vector-less testing with JTAG boundary scan. JTAG allows vector-less testing of the PCBs using only the five universal end ports to test these components. So, instead of having the higher number of test points you can test those components with those five points.

Assemblies are getting more complex day by day; you cannot have the test point on each and every point. Moreover, all the complex components in the latest PCBs are boundary-scan enabled. In this context, methods like JTAG boundary scan make it more convenient to access the I/O pins of the JTAG-compliant ICs present on the circuit board.

Even if you have a multiple chain of components that are boundary-scan enabled, these can be tested all together in one chain. This ability to daisy chain all the JTAG-compliant compo-

Key advantages of integrated boundary scan

- **Extended test channel.** Taking an example, if you have a 50-pin device, you can drive the boundary scan devices through the JTAG ports, which are only five-pin ports, and the 50-pin device now becomes your extended test channel.
- **Bi-directional testing.** Apart from the forward direction, you can also test in the reverse direction, that is, from the edge connector till the boundary scan device.

Design for testability (DFT)

According to the experts, selection of a suitable ICT system for your product depends upon the type of your product. A design should have some testability associated with it. DFT adds certain testability features to the product design, making it easier to develop and apply manufacturing tests for the designed hardware.

Ravi explains, "We study what DFT the board is having. For example, it can be a network board or a communication board. But first we have to study what kind of components are used, and whether there is any DFT or not. If it does not have DFT, we have to make the analysis and then ask the design team to get that assembly ready for testing. If there is no DFT then no testing can be done."

nents on the circuit board, reduces the amount of wiring while keeping similar test coverage. Apart from the vector-less testing you can also perform on-board programming of the components using JTAG ports.

High-density channel cards with hybrid test points. But now we have single integrated systems with high-density channel cards, combining various test flows into a single, full-test performance insertion. These can dramatically reduce your production test times, recurring test costs and improve time to market significantly.

The new LX2 system from Teradyne having 128 channel cards can test complex boards which require up to 15,360 hybrid test points in a single insertion. It has full analogue, digital, boundary scan and in-system programming capability. Each 128HD card delivers 512 hybrid test pins, doubling test system pin counts of the existing systems.

Parametric testing. Parametric measurement units (PMUs) are integrated with the latest ICT systems. Usually these come with 4 PMUs, capable of measuring voltage and current with respect to reference current and voltage, respectively. This makes them very use-